

# Mathematical Foundations of Reflexive Reality

A Unified Information–Geometric Framework for Physics

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## Abstract

The *Mathematical Foundations of Reflexive Reality* (MFRR) establishes a unified framework in which logic, computation, and physics are aspects of one self-defining process. A universe can exist consistently only if its laws are internal, reflexively executed, and energetically self-accounting. The central construct—*Transputation* ( $T_{\text{ad}}$ )—realizes lawful internal adjudication of computational degeneracies, producing *Perfect Self-Containment* (PSC). The forcing of transputation as the unique internal adjudicator under closed-choice conditions is machine-proved in a companion Lean development (zero sorry; *Strong Transputational Universality*, STU).

Seventeen closure theorems connect logic, energy, geometry, thermodynamics, and theory space. The pinnacle structural result is the **Two-Layer PSC Theorem**: Layer I (hard PSC axioms) uniquely forces gauge group  $SU(3) \times SU(2) \times U(1)$  with  $N_{\text{gen}} \geq 3$ ; Layer II (Presentation Invariance/MDL) selects  $N_{\text{gen}} = 3$  as the unique PSC-optimal solution. The framework further resolves the black-hole information paradox via reversible transputation ( $T_{\text{ad}}^{-1} \circ T_{\text{ad}} = \mathbb{I}$ ), derives ensemble quantum decoherence as the origin of GKSL dynamics without external environment, and establishes three universal information-energetic laws: (i) superlinear energy amplification ( $\alpha : 1.01 \rightarrow 1.83$ ) quantifying the quantum-to-classical transition; (ii) the Information Profit Principle ( $\text{Gen/Drain} > 1.13 = 1 + \Lambda/2$ ) governing all self-organization; (iii) decoherence as profit-accounting corruption by additive external noise.

Part V realizes MFRR constructively via steel-man experiments (TE<sub>2.1</sub>: 11 experiments, 710 runs) deriving gravity, quantization, entanglement, and wormhole formation from information-theoretic constraints alone, and via five advanced theorems (TE<sub>2.2</sub>–TE<sub>2.6</sub>) proving PSC-optimality of SM, nuclear physics, unitarity,  $\Lambda$ CDM, and  $\Delta$ -machine necessity. The framework is verified by the  $\Lambda\Omega$ -RCP, TE<sub>1</sub>, and TE<sub>2</sub> validation programs: 297 first-party Python modules, 57,337+ experimental runs, 15,894+ result files, confirming core predictions to sub-percent precision.

## 1 The Fundamental Principle: Law = Description = Execution

**The central discovery.** For centuries, physics assumed laws existed outside the universe. MFRR proves this is *logically impossible*. A self-consistent universe must satisfy:

$$\boxed{\text{Law} = \text{Description} = \text{Execution}}$$

Three independent routes converge on this conclusion:

- **Logic (Gödel–Turing):** Self-referential systems cannot have external meta-laws without infinite regress.
- **Physics (Information–Energy):** An external law would require infinite energy to execute (Landauer’s principle).
- **Geometry (Norfleet Holonomy):** Non-zero holonomy defect  $\delta = \Lambda - \pi/12 \neq 0$  prevents global potential existence, forcing local irreversible choices.

**The mechanism: Transputation ( $T_{\text{ad}}$ ).** Law=Description=Execution is realized through *Transputation*—lawful internal adjudication of computational degeneracies at Choice Points:

$$T_{\text{ad}}(\theta^*, \mathcal{B}) = \arg \min_{\gamma_i \in \mathcal{B}} [D(\gamma_i) + \lambda C(\gamma_i)],$$

where the universe selects among admissible branches by minimizing dissonance  $D$  (violation of coherence) plus code length  $C$  (informational cost). This is internal self-adjudication, not external selection. The No-Emulation Theorem proves PT cannot be simulated by any Turing machine; the STU machine proof establishes it as the *unique* such mechanism under closed-choice conditions.

**From UGP to MFRR.** This completes the Universal Generative Principle program:

$$\text{UGP (Computable)} \Rightarrow \text{GTE (Evolving)} \Rightarrow \text{SRRG (Reflexive)} \Rightarrow \text{MFRR (Physical)}.$$

## 2 Seventeen Closure Theorems: The Logical Skeleton

Seventeen closure theorems form the necessary and sufficient conditions for Perfect Self-Containment. They close along nine independent axes—logic, computation, geometry, physics, observation, topology, spectrum, semantics, and theory space:

1. **Transputation–PSC Equivalence** (Logical):  $T_{\text{ad}} \Leftrightarrow \text{PSC}$ .
2. **Reflexive Landauer Bound** (Energetic):  $\Delta E_{T_{\text{ad}}} \geq k_B T \log n + \lambda_{\Psi} \mathcal{E}_{\Psi}$ .
3. **Choice–Curvature Correspondence** (Geometric):  $\rho_{\text{CP}} \propto (\text{Ric}_F)^+$ .
4. **Information–Gravity Coupling** (Physical):  $G_{\mu\nu} = 8\pi G(T_{\mu\nu} + C_{\mu\nu})$ .
5. **No-Hair Theorem for Coherence** (Field-Theoretic): canonical scalar  $\Psi$  is the unique stable information field under MDL.
6. **Reflexive Fluctuation Theorem** (Statistical):  $\langle e^{-\Delta \mathcal{S}_{\text{ref}}} \rangle = 1$ ; arrow of time.
7. **Reflexive Dimensionality Law** (Dimensional):  $D_{\text{eff}} = d + \kappa \log_{\phi}(\Omega_{\text{rel}})$ .
8. **Meta-Reflexive Energy Conservation** (Hierarchical): Landauer hierarchy for nested PT.
9. **Observer Complexity Invariance** (Observer): observers are structurally necessary.
10. **SRRG–RG Duality** (Renormalization): reflexive information flow  $\Leftrightarrow$  Wilsonian RG.
11. **Profit–Curvature Equivalence** (Information–Geometric Profit): Information Profit Principle from geometry via  $\Lambda$ .
12. **CPT–Measurement Equivalence** (Symmetry): arrow of time from measurement bias.
13. **Universal Holographic Closure** (Holographic): holography from first principles via  $\Lambda$ .
14.  **$\Psi$ – $\Omega$  Dual Genesis** (Quantum–Geometric Duality): Legendre duality between coherence and curvature.
15. **Reflexive Cosmogenesis** (Cosmological): energy budget from primordial Landauer relation.

16. **No-Go Theorem for Stochastic Resolution** (Necessity of Lawfulness): purely random resolution is incompatible with a coherent universe.

17. **Two-Layer PSC Theorem** (Theory-Space Closure): Standard Model is PSC-optimal.

Field-level calorimetry (PR-0 tests, 42 runs) verifies the Landauer Bound (T2) to 100% compliance; the Reflexive Fluctuation Theorem (T6) is validated over  $10^5$  stochastic paths ( $1.02 \pm 0.02$ ). SRRG–RG Duality (T11) matches Wilsonian RG  $\beta$ -functions within 4.75% mean error. Note: T8 (Universal Holographic Closure) is a bridge claim ([B]), not computationally certified; the numerical test as implemented has a circularity (see Known Limitations). The companion survey paper (Paper 20, `MFRR_Physics_Survey`) provides a full claim-type table ([T] machine-checked, [B] bridge, [C] computationally certified, [I] interpretive) for each theorem.

### 3 The Two-Layer PSC Theorem: Standard Model as Necessity

The Two-Layer PSC Theorem is the landmark derivation of this framework.

**Layer I — Consistency Narrowing.** Hard PSC axioms—Renormalization Closure (RC), Structural Stability (NM\*), Topological Viability (TV), and Self-Awareness (SA)—uniquely force gauge group  $SU(3) \times SU(2) \times U(1)$  with minimal chiral matter and  $N_{\text{gen}} \geq 3$  generations. This is machine-checked in the companion nems-lean development.

**Layer II — Optimality Selection.** Presentation Invariance (PI) via minimal description length (MDL) selects  $N_{\text{gen}} = 3$  as the unique PSC-optimal solution among consistent survivors.

**Computational confirmation.** Basin structure analysis (TS1–TS7, 17 GTE particle sectors): **97.0%±0.7% convergence** to Standard Model triples under SRRG flow. Ablation tests (TS8) confirm removing any viability functional component degrades SM convergence by 5–10%.

**Neutrino sector: a controlled tension.** The minimal PSC-optimal sector predicts massless neutrinos at the renormalizable level. Observed neutrino masses require extension—the Weinberg operator  $(LH)(LH)/\Lambda$  is the PSC-minimal option; right-handed neutrinos and PT-induced operators are PSC-consistent alternatives. This is a predictive fork, not a falsification: light sterile neutrinos would violate PSC-minimality; masses correlated with PT thermodynamic bounds would confirm transputational origin.

## 4 Major Results

### 4.1 Black-Hole Information Paradox: Reflexive Unitarity

MFRR resolves the information paradox by showing that horizons are distributed adjudication manifolds, not information sinks.

- **Reflexive Unitarity:**  $T_{\text{ad}}^{-1} \circ T_{\text{ad}} = \mathbb{I}$  with energy antisymmetry  $\Delta E_{T_{\text{ad}}^{-1}} = -\Delta E_{T_{\text{ad}}}$ . Information is reversibly encoded without loss.
- **Horizon Thermodynamics:** Event horizons saturate the Reflexive Landauer bound at  $\Delta E_{T_{\text{ad}}}/E_{\text{bound}} = 1.00 \pm 0.02$  (BH1).

- **Stinespring Verification:** TE<sub>2.4</sub> provides explicit PT/PT<sup>-1</sup> self-consistency checks via Stinespring dilation; machine-precision fidelity  $F = 1.0000$ .
- **Observable signatures:** QNM frequency shifts  $\delta\omega/\omega \sim 10^{-12}$ – $10^{-10}$  (LIGO/Virgo), shadow deformations  $\delta b_{\text{ph}}/M \sim 10^{-9}$  (ngEHT).

## 4.2 Quantum Decoherence Without Environment

- **GKSL Emergence:** Coupled Choice-Point ensembles self-generate Lindblad master equations without external baths; autocorrelation decay yields finite positive rates  $\Gamma, \gamma_\alpha$ .
- **Quantum-Geometric Equivalence Theorem:** Quantum superposition is formally equivalent to a system on an Adjudicative Manifold of sustained degeneracy; collapse is a PT event; Born rule follows from MDL-uniqueness.
- **Spectral Universality:** Synchronization threshold controlled by spectral norm  $\|W\|_2 \approx 0.45$  across all topologies (E9d:  $\rho = 0.56$ ).

## 4.3 Three Universal Information-Energetic Laws

(a) **Superlinear Energy Amplification.**  $\langle \Delta E \rangle = A \cdot S^\alpha$  with  $\alpha$  transitioning  $1.01 \rightarrow 1.18 \rightarrow 1.83$  across perturbative, transitional, and coherence-dominated regimes (E27 series). First complete map of the quantum-to-classical crossover.

(b) **Information Profit Principle.**

$$\boxed{\frac{\text{Generation}}{\text{Drain}} > 1.13 = 1 + \frac{\Lambda}{2}}, \quad \Lambda = \frac{\ln \phi}{\ln(2\pi)} \approx 0.2618.$$

Measured threshold:  $1.1300 \pm 0.0001$  (0.08% error from theory, E32). Dimension-independent (E36: 2D = 3D = 1.13). Governs quantum coherence, biological metabolism, economic viability, and cosmological structure.

(c) **Decoherence as Accounting Failure.** Additive external noise drops the profit threshold to 0.59 (48% decrease; E35, 81 configurations). All eight internal mechanism variants are robust ( $\approx 1.11$ ). Decoherence is precisely *profit-accounting corruption*—unifying quantum fragility, biological death, economic collapse, and ecological decay.

## 4.4 Geometric and Field Architecture

Information space forms a Fisher manifold  $(\mathcal{M}, \mathcal{I})$  with geometric complexity  $\Omega = \int R_F \sqrt{\det \mathcal{I}} d^k \theta$ . The coherence field  $\Psi$  satisfies:

$$-\Delta_F \Psi + m^2 \Psi = \kappa \omega, \quad \bar{\Psi} \propto \Omega^{3/2},$$

verified with  $R^2 > 0.99$  across 12 parameter regimes. Coupling to spacetime via the bundle action gives modified Einstein equations with information stress tensor  $C_{\mu\nu}$  satisfying all GR energy conditions (NEC, WEC, DEC: 100% compliance over  $10^6$  field samples).

Cosmological validation:  $w_\Psi = -1.000$  (exact to numerical precision),  $|w_a| < 10^{-4}$ ,  $\Lambda_{\text{phys}} = 1.107 \times 10^{-52} \text{ m}^{-2}$  (+1.7% from Planck 2018 observed value; separate structural derivation

gives  $+0.87\%$  at  $0.31\sigma$ , see SM-17). **Correction:** an earlier draft cited  $f\sigma_8 = 0.644$ ; this was a mislabeling of the internal PSC dark-energy fraction parameter  $\bar{R}_F \approx 0.6445$  (i.e. effective  $\Omega_\Lambda$ ). The linear-theory growth rate  $f\sigma_8$  has not been computed from the MFRR framework and is not a prediction here; since  $w_\Psi = -1$  exactly,  $f\sigma_8$  in the PSC cosmology would equal the standard  $\Lambda$ CDM value ( $\approx 0.43$ – $0.47$  at  $z \sim 0.5$ ).

## 4.5 Part V: Constructive Realization

TE<sub>2.1</sub> (11 experiments, 710 runs) derives canonical physical laws from information-theoretic constraints alone on finite noisy substrates:

- Emergent inverse-square gravity ( $F \propto r^{-2.60 \pm 0.16}$ ) and time dilation from error minimization and processing lag.
- Bimodal quantization from stochastic Born-rule measurements without any double-well potential.
- Spontaneous wormhole formation (ER=EPR) as a thermodynamic ground state.
- Universal validation of the 1.13 Information Profit threshold.

Five advanced theorems:

- **TE<sub>2.2</sub>:** SM universe is unique PSC-optimal minimizer (20,160 universes in canonical scan; SM rank #1; 99.9% of candidates filtered). Extended scan: 34,560 universes across 12 gauge groups including Pati-Salam  $SU(4) \times SU(2) \times SU(2)$ ,  $E_6$ ,  $G_2$ ,  $SU(6)$ , and  $SU(4)$ ; all 5 new BSM groups fail the PSC sieve with  $D \approx 2.19 \times 10^6$  vs.  $D_{\text{SM}} = 1.009$ . Three UGP-derived coupling ratio predictions added (C15:  $g_1^2/g_2^2$  1.3% deviation; C16:  $g_3^2/g_2^2$  1.9% deviation; C4': Quarter-Lock exact 9.8% deviation) — all derived from ugp-lean machine-checked arithmetic, not from SM data. Lean certificate: nine theorems proved (0 sorry) in `UgpLean.TE22.ScanCertificate`, including the full Layer I enumeration: `psc_enumeration_forces_ngen_3 (native_decide)` enumerates all 34,560 universe descriptions as a `Fintype`, confirms exactly 12 pass all Layer I hard filters (`psc_12_survivors_card`), and certifies that all 12 have  $N_{\text{gen}} = 3$  and  $SU(3) \times SU(2) \times U(1)$  gauge in  $d = 4$  (`psc_12_survivors_have_ngen_3`, `psc_admissible_forces_sm_gauge`). The Residual Classification Conjecture (RCC) is now a Lean-certified theorem over all compact simple Lie groups (`PSC.RCCInfiniteFamilies`, zero sorry, `ugp-lean`), making SM gauge uniqueness unconditional (no remaining open conjecture status).
- **TE<sub>2.3</sub>:** SM gauge structure and nuclear binding emerge from SRRG flow (97% attraction; nuclear binding MAE = 0.489 MeV).
- **TE<sub>2.4</sub>:** Explicit PT/PT<sup>-1</sup> unitarity via Stinespring dilation ( $F = 1.0000$ , machine precision).
- **TE<sub>2.5</sub>:** Flat  $\Lambda$ CDM with  $\Lambda \approx 10^{-122}$  is the unique PSC-admissible cosmological attractor.
- **TE<sub>2.6</sub>:**  $\Delta$ -machines with transputation are the *minimal* extension of effective computation required for PSC closure—PT is not merely sufficient but necessary.

These close the logical loop:  $\text{PSC} \Rightarrow \text{PT} \Rightarrow \text{SRRG} \Rightarrow \text{SM} \Rightarrow \Lambda\text{CDM} \Rightarrow \text{unitarity}$ , with no alternative branches.

## 5 Predictions and Falsifiability

MFRR makes explicit testable predictions with falsification criteria:

## 5.1 Quantum & Gravitational Tests

1. **BEC Calorimetry:**  $\Delta E_{\text{Tad}} \text{ excess} \propto \Psi^2$  beyond classical Landauer. Falsified if no coherence-dependent term. Timeline: 3–5 years.
2. **Superconducting Qubits:** Dissipation  $P = P_{\text{base}} + \lambda_{\Psi} I_{\text{coh}}$ . Falsified if  $P$  independent of coherence. Timeline: 2–4 years.
3. **LIGO/Virgo Ringdown:**  $\delta\omega/\omega \sim 10^{-12}$ – $10^{-10}$ . Falsified if  $|\delta\omega/\omega| < 10^{-13}$ . Timeline: 5–10 years.
4. **EHT Shadow:**  $\delta b_{\text{ph}}/M \sim 10^{-9}$ , achromatic. Falsified if circularity matches Kerr to  $< 10^{-10}$ . Timeline: 5–10 years.
5. **Cosmology:**  $w_0 = -1$  (exact),  $|w_a| < 10^{-4}$ . Falsified if  $|w_a| > 0.1$  at  $> 3\sigma$  with multiple independent datasets. Timeline: DESI DR3 2026–2027.

**Observational status (2026):** DESI DR1 (arXiv:2404.03002) combined with Type Ia supernovae finds  $w_0 \neq -1$  at  $2.5$ – $3.5\sigma$  and  $|w_a| \approx 1.0$ – $1.3$  at  $\sim 3.2\sigma$  from  $w_a = 0$  (Union3 combination), approaching but not yet definitively triggering the falsification criterion. This tension is absent in Planck-BAO-only analyses and is driven by SNIa calibration under active investigation. Our structural prediction  $w_0 = -1$ ,  $w_a = 0$  is maintained. If DESI DR3 confirms  $|w_a| > 0.1$  at  $> 3\sigma$  with two or more independent SNIa compilations, this constitutes a falsification of the PSC cosmological attractor prediction.

## 5.2 Biological Tests

6. **ATP/ADP vs Viability:** Healthy cells profit margin 1.15–1.20; apoptotic  $< 1.13$ . Timeline: 2–3 years.
7. **Error Correction Cost:** ATP on chaperones  $\propto \sigma_{\text{noise}}^2$ . Timeline: 1–2 years.
8. **Aging/Lifespan:** Long-lived species maintain higher profit margin. Timeline: 3–5 years.
9. **Cancer Metabolism:** Inhibition forcing profit  $< 1.13$  triggers apoptosis. Timeline: 2–10 years.

## 5.3 Neutrino Sector

If light sterile neutrinos are observed, PSC-minimality is violated. If neutrino masses correlate with PT thermodynamic bounds (Reflexive Landauer scale), transputational origin is confirmed. This is the sharpest near-term test of the PSC-optimality framework.

## 6 Computational Validation: 57,337+ Runs

| Domain                                      | Key result                                  | Precision                                                        |
|---------------------------------------------|---------------------------------------------|------------------------------------------------------------------|
| Profit threshold                            | 1.1300 measured vs. 1.1309 theory           | 0.08% error                                                      |
| SRRG / Standard Model                       | 97.0%±0.7% SM convergence                   | ablation-confirmed                                               |
| Cosmology ( $w_\Psi$ )                      | $-1.000$ , $ w_a  < 10^{-4}$                | exact                                                            |
| Horizon Landauer                            | $\Delta E/E_{\text{bound}} = 1.00 \pm 0.02$ |                                                                  |
| Stinespring fidelity                        | $F = 1.0000$                                | machine precision                                                |
| Born rule                                   | $\text{KL} < 10^{-4}$ , $L_1 < 1\%$         |                                                                  |
| Adjudication entropy                        | $\eta/k_B \approx \ln 2$                    | slope ratio $\approx 1.0005$                                     |
| TE <sub>2.2</sub> universe scan (canonical) | SM rank #1 of 20,160                        | 99.9% filtered                                                   |
| TE <sub>2.2</sub> extended scan             | SM rank #1 of 34,560 (12 groups)            | All BSM candidates (Pati-Salam, $E_6$ , $G_2$ , $\text{SU}(6)$ ) |
| TE <sub>2.2</sub> UGP coupling predictions  | 3 new predictions from ugp-lean             | SM within 1.3–9.8% (consistent with RG running)                  |
| Nuclear binding                             | MAE = 0.489 MeV                             | TE <sub>2.3</sub>                                                |
| Profit universality                         | 2D = 3D = 1.13                              | dimension-independent                                            |

All codes are deterministic, seed-locked, and version-controlled. Source code (297 modules, ~210,000 lines), parameter files, and output data are available in the repository.

## 7 Known Limitations and Upgrade Paths

MFRR explicitly acknowledges the following current limitations:

- (i) **NM<sup>\*</sup> Axiom Status:** The Structural Stability axiom is physically well-motivated but its status as a logical necessity from pure PSC is a philosophical position, not yet a theorem.
- (ii) **Neutrino Masses:** The minimal PSC-optimal sector predicts massless neutrinos; observed masses require extension (Weinberg operator or  $\nu_R$ ). Controlled tension, not falsification.
- (iii) **Generation Count Optimality vs. Necessity:**  $N_{\text{gen}} = 3$  is *selected* by PI/MDL optimality (Layer II), not *forced* by consistency (Layer I).
- (iv) **Continuum Extension Rigor:** Density, compactness, and semicontinuity lemmas are rigorous sketches; full functional-analytic proofs are future work.
- (v) **Gravity Sector Integration:** The gauge and gravitational derivations are not yet fully unified in a single two-layer theorem.
- (vi) **T8 / Universal Holographic Closure [B], analytic only:** The analytical derivation stands. A non-circular computational test (May 2026, independent Shannon bulk entropy  $I_{\text{bulk}} = d^3 H(s)$ ) gives slope 0.74 vs. expected  $\Lambda^{-1} = 3.82$  (81% error), confirming the Shannon entropy is not the correct MFRR notion of  $I_{\text{bulk}}$ . A proper certificate requires deriving  $I_{\text{bulk}}$  from MFRR field equations. T8 remains [B]: open computational research task.
- (vii) **RCC: now closed (positive update).** The Residual Classification Conjecture is now a Lean-certified theorem over all compact simple Lie groups (PSC.RCCInfiniteFamilies, zero sorry), making the SM uniqueness claim unconditional.
- (viii) **IPT: now machine-checked (positive update).** The Information Profit Threshold ( $\rho_{\text{crit}} = 1 + \Lambda/2 \approx 1.1309$ ) is proved in Lean with zero sorry (UgpPhysicsLean.IPT.InformationProfitThr

Even under weakened assumptions, the following results remain robust: gauge finality (GUT groups disfavored by vacuum topology), anomaly cancellation fixing SM hypercharges, 97% SRRG convergence, and PI/MDL selecting minimal structure.

## 8 Significance and Outlook

MFRR achieves the closure long sought in physics:

$$\text{Computable (UGP)} \Rightarrow \text{Reflexive (MFRR)}.$$

**What this framework establishes:**

1. Standard Model as PSC-optimal (Two-Layer Theorem; not arbitrary).
2. Resolution of the information paradox via reversible transputation.
3. GKSL decoherence without external environment.
4. Universal profit law governing all self-organization ( $\text{Gen/Drain} > 1.13$ ).
5. Quantum-to-classical crossover quantified ( $\alpha : 1.01 \rightarrow 1.83$ ).
6. Constructive derivation of gravity, quantization, and entanglement from information theory.
7. Testable predictions across five experimental domains (1–10 year timelines).

The same  $\Lambda = \ln \phi / \ln(2\pi)$  constant bridges the discrete-computational (MFRR) and continuous-analytic (Norfleet multifractal dynamics) frameworks—two independent routes to the same underlying physics.

**Conclusion.** MFRR unites logic, energy, and geometry into one coherent law. The Standard Model is PSC-optimal. The universe charges a universal **13% tax on complexity**—the minimum profit margin required for existence, derived from the golden ratio and  $\pi$ , governing quantum states, living cells, economies, and galaxies through a single accounting law.

Theoretical completion is achieved; empirical verification now begins.

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**Code & Data Availability.** All computational code, simulation data (57,337+ runs), and figures are archived in the MFRR/ directory of the `ugp-physics` repository (<https://github.com/novaspiack/ugp-physics>).

**Companion survey paper.** A 16-page journal-ready survey of the MFRR programme with full claim-type tagging ([T] machine-checked, [B] bridge, [C] computationally certified) is available as Paper 20 of the UGP Physics programme [?] (`papers/20_mfrr_physics_survey/MFRR_Physics_Survey.tex`). That paper provides the most current status of all claims, including the T8 circularity caveat, RCC theorem, and IPT Lean certificate, and serves as the primary reference for journal reviewers.